Design and Low-Power Analysis of an 8-Bit PIPO Register Using Shared and Fixed 4-Bit Grouped Data-Driven Clock Gating

Sai Ram Ruttala

Under the Guidance of

Dr. Paritosh D. Peshwe

Department of Electronics and Communication Engineering

Indian Institute of Information Technology Nagpur, Butibori,

441108, India

Abstract

Clock activity is an important contributor to dynamic power in synchronous digital circuits. A register may receive clock transitions even when the data stored in it does not change. Data-driven clock gating (DDCG) attempts to reduce unnecessary clock activity by comparing incoming data with the currently stored output and enabling clock delivery only when a change is detected. This paper studies an 8-bit parallel-in parallel-out (PIPO) register using two DDCG organizations: a shared architecture with one clock-gating control group for all eight bits, and a fixed 4-bit grouped architecture with two independent control groups. The supplied Microwind power readings are 188.336 mW for shared DDCG and 175.422 mW for grouped DDCG. The absolute difference is 12.914 mW, equivalent to a 6.8569% reduction relative to the shared result. This comparison is limited to the two reported readings; it does not establish power savings relative to a conventional PIPO register because that baseline measurement has not yet been supplied. The paper also discusses four 2-bit groups and individual per-bit gating for an 8-bit register, then extends the structural grouping analysis to 16-bit and 32-bit registers. An illustrative switching-probability model explains why smaller groups can reduce the number of flip-flops receiving clock pulses, while additional gating overhead can offset that benefit. No absolute power values are extrapolated to wider registers. The results and discussion are intended as a low-power-focused B.Tech-level study, with assumptions and pending verification stated explicitly.

Keywords: low-power VLSI, data-driven clock gating, PIPO register, clock gating, flip-flop grouping, switching activity

1. Introduction

Registers are fundamental elements in processors, embedded controllers, communication systems, and digital signal-processing circuits. In a synchronous register, a clock signal controls when input data is captured. Because the clock repeatedly transitions, clock networks and clock pins can consume dynamic power even when the stored data is unchanged. Clock gating reduces clock activity by preventing pulses from reaching parts of a circuit that do not need to update.

Conventional enable-based clock gating commonly uses a functional enable signal to indicate whether a register must operate. Data-driven clock gating uses the relationship between input data and stored output data to detect whether an update is needed. For an N-bit register, a change detector can be written as $C_i = D_i \oplus Q_i$. The change signals can be combined into one enable for the entire register or separated into independent groups. Grouping is important because a larger group uses fewer gating circuits but may clock more flip-flops than necessary. Smaller groups can avoid more unnecessary clock pulses, but require more gating circuitry and routing [1], [2].

This work studies a small 8-bit PIPO register in two organizations for which power readings are available: shared DDCG and fixed 4-bit grouped DDCG. Additional grouping options are analyzed to explain possible low-power trade-offs, not to imply that their power has already been measured.

1.1 Problem statement

In the shared architecture, a change in any bit may enable the clock for the entire 8-bit word. In the fixed 4-bit grouped

architecture, a change in one half-word can activate the corresponding group while the other group may remain clock-disabled. The design question is whether the clock-activity reduction from grouping is large enough to offset the extra clock-gating and control overhead.

1.2 Objectives

Compare the supplied power readings of shared and fixed 4-bit grouped DDCG for the 8-bit register.

Explain the DDCG operating principle and the need for glitch-free clock gating.

Discuss the low-power trade-offs of one shared group, two 4-bit groups, four 2-bit groups, and per-bit gating.

Provide structural grouping counts for 16-bit and 32-bit register designs without inventing absolute power measurements.

Identify the additional experimental information required for a reproducible low-power comparison.

1.3 Scope and contribution

The measured comparison in this paper is restricted to the two power values supplied for the 8-bit shared and fixed 4-bit grouped implementations. A measured power value for the conventional PIPO reference is not available in the current result set. Accordingly, this paper does not calculate an improvement over the conventional register, does not analyze physical area or propagation delay, and does not claim that fixed 4-bit grouping is universally optimal. The contribution is a careful low-power comparison of these implementations with an analytical discussion of grouping choices.

2. Circuit Design and Low-Power Principle

2.1 Conventional 8-bit PIPO register

An 8-bit PIPO register accepts D[7:0] and stores it in eight flip-flops. The outputs Q[7:0] represent the stored word. At each active source-clock edge, the register captures the input vector when the clock is delivered. The conventional circuit has no data-change detection logic. It is included as the design reference, but its power measurement must be obtained before making a measured comparison against DDCG.

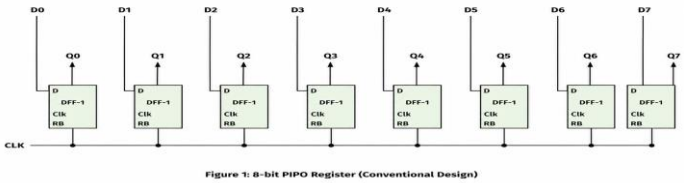


Fig. 1. Conventional 8-bit PIPO schematic.

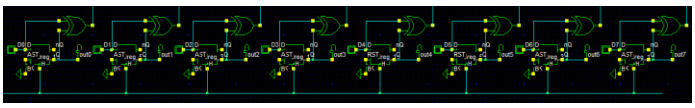


Fig. 2. Conventional 8-bit PIPO architecture.

2.2 Bit-level data-change detection

Each bit uses an XOR function to compare the input with the stored output:

$$C_i = D_i \oplus Q_i, \quad i = 0,1,\dots,7$$

If D_i and Q_i are equal, C_i = 0. If they differ, C_i = 1. The change signals are combined within a group to produce an enable. A clock-gating circuit uses that enable to determine whether the group receives a valid source-clock pulse. For positive-edge-triggered registers, the enable should be captured while CLK is low and held stable while CLK is high, for example through a suitable latch-based ICG implementation. The timing and pulse behavior must be verified in simulation.

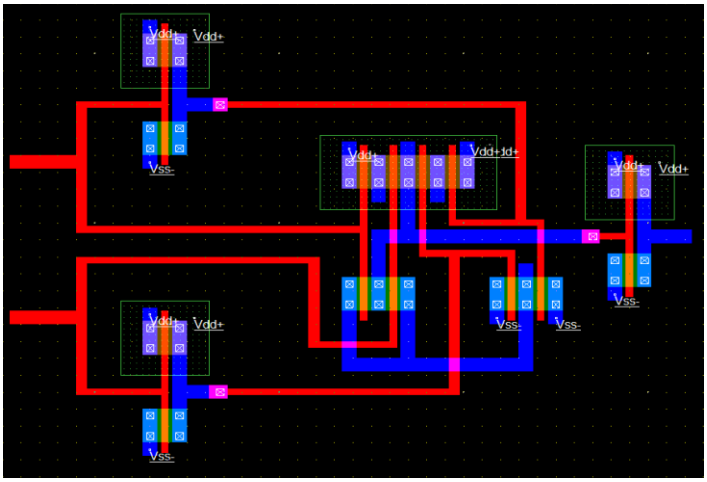


Fig. 3. XOR change detector and glitch-free clock-gating cell.

2.3 Shared DDCG architecture

The shared DDCG architecture uses eight XOR change detectors. Their outputs are combined into one shared enable:

$$E_{\text{shared}} = C_0 \vee C_1 \vee C_2 \vee C_3 \vee C_4 \vee C_5 \vee C_6 \vee C_7$$

One ICG drives the gated clock for all eight flip-flops. This arrangement minimizes the number of independent gating circuits. However, if only one bit changes, the entire 8-bit register may receive a clock pulse. The shared architecture therefore trades simple, low-overhead gating against the possibility of clocking bits that do not need to change.



Fig. 4. Shared DDCG layout.

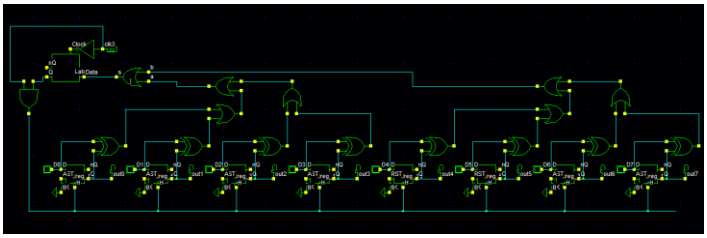


Fig. 5. Shared DDCG architecture.

2.4 Fixed 4-bit grouped DDCG architecture

The fixed grouped architecture divides the register into two 4-bit groups. Group 0 contains bits 0-3 and Group 1 contains bits 4-7. Each group has its own reduction logic and ICG:

$$E_{\text{low}} = C_0 \vee C_1 \vee C_2 \vee C_3$$
$$E_{\text{high}} = C_4 \vee C_5 \vee C_6 \vee C_7$$

Both ICGs receive the same source clock, but their outputs drive separate sets of four flip-flops. If a change occurs only in the lower group, the upper group need not receive a clock pulse, provided the gating implementation and timing behave as intended. This can reduce clock switching, while the second ICG adds power overhead of its own.

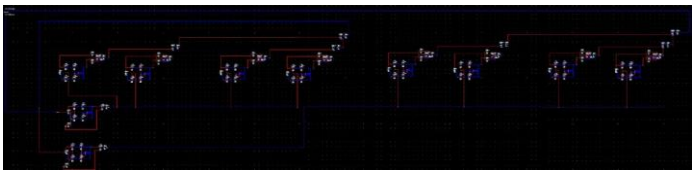


Fig. 6. Fixed 4-bit grouped DDCG layout.

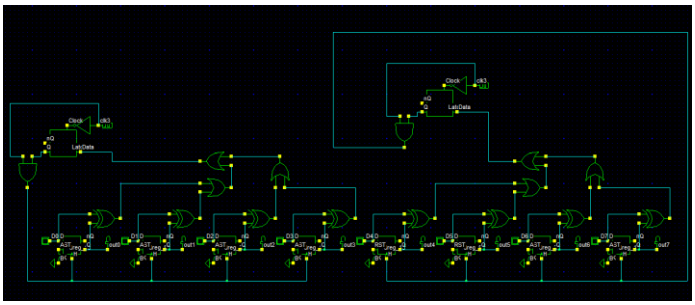


Fig. 7. Fixed 4-bit grouped DDCG architecture.

3. Low-Power Evaluation Methodology

3.1 Reported power results

The only experimental values currently supplied for the 8-bit DDCG comparison are the following Microwind power readings. They are retained exactly as reported:

Architecture	Reported power	Status
Shared DDCG (one group of 8 bits)	188.336 mW	User-supplied reading
Fixed 4-bit grouped DDCG (two groups)	175.422 mW	User-supplied reading
Conventional 8-bit PIPO	Not supplied	Measure under the same conditions

The exact label used by Microwind for the displayed power metric (for example, average power or total power) should be copied into the final report. Unless the tool readout explicitly identifies the measurement as dynamic power, the values should be described as reported power readings rather than relabelled as dynamic power.

3.2 Conditions required for a fair comparison

A reliable power comparison requires both designs to use the same process rule file, supply voltage, source-clock frequency and duty cycle, input sequences, initialization/reset conditions, simulation interval, output loading, and power metric. The same measurement window should be used after allowing any startup conditions to settle. If these conditions were not identical in the original runs, repeat the measurements before making a strong causal claim.

Record the exact Microwind version and technology rule file.

Record VDD, clock frequency, duty cycle, and the exact reported power metric.

Apply the same input pattern sequence to the shared and grouped circuits.

Use the same initial Q state, reset procedure, simulation duration, and output load.

Save screenshots of each result window and the relevant input/output waveforms.

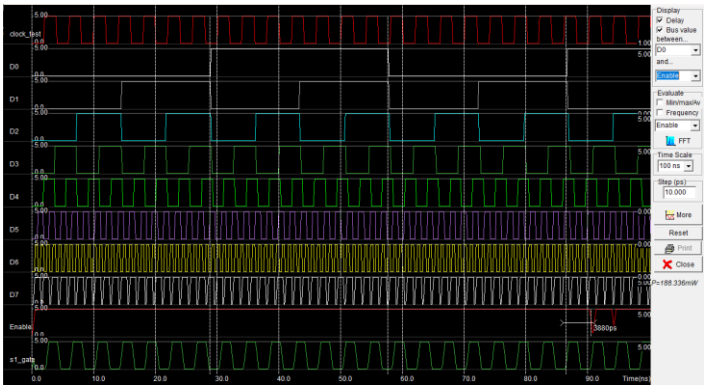


Fig. 8. Shared DDCG power result in Microwind.

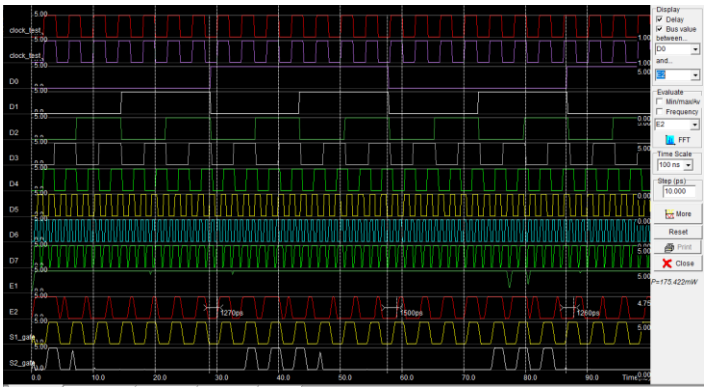


Fig. 9. Fixed 4-bit grouped DDCG power result in Microwind.

3.3 Power calculation

The absolute difference between the two supplied readings is:

$$\Delta P = P_{\text{shared}} - P_{\text{grouped}} = \mathbf{188.336 - 175.422 = 12.914mW}$$

The relative reduction with the shared design as the reference is:

$$\text{Reduction (\%)} = [(P_{shared} - P_{grouped}) / P_{shared}] \times 100$$

$$= (12.914 / 188.336) \times 100 = 6.8569\%$$

The percentage describes the difference relative to the shared DDCG reading. It does not establish the percentage power savings relative to the conventional register, and it should not be generalized to wider registers without new measurements.

4. Results and Discussion

4.1 Measured comparison

Parameter	Shared DDCG	Fixed 4-bit grouped DDCG
Register width	8 bits	8 bits
Number of independently gated groups	1	2
Reported power	188.336 mW	175.422 mW
Absolute difference vs. shared	Reference	-12.914 mW
Relative power reduction vs. shared	Reference	6.8569%

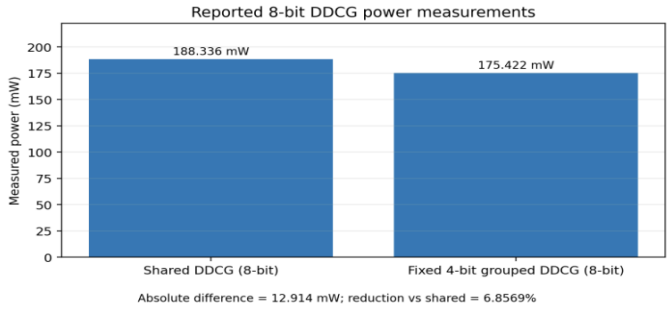


Fig. 10. Power readings reported for the 8-bit shared and fixed 4-bit grouped DDCG circuits. Values are those supplied for this study.

The fixed 4-bit grouped implementation has a lower reported power value than the shared implementation by 12.914 mW. One plausible explanation is that the grouped design can keep the unaffected half-word clock-disabled when only the other half-word changes. This benefit is not guaranteed for every stimulus: if both groups frequently change together, the independent gating may provide less reduction while the extra ICG continues to consume power.

The measured difference is consistent with the idea that reducing unnecessary clock delivery can reduce power, but it does not by itself prove the cause of the reduction. Confirmation depends on equivalent measurement conditions and correct glitch-free gating. Further tests should separate low-group-only activity, high-group-only activity, changes in both groups, and no-data-change intervals.

5. Low-Power Trade-Offs of Different Group Sizes

The group size sets a trade-off between selectivity and gating overhead. A single shared group needs only one ICG, but any bit change can clock the whole word. Dividing the word into smaller groups increases selectivity: only the groups containing changed bits need clock pulses. The cost is a larger number of ICGs, enable paths, gated-clock branches, and routing connections. Since each ICG consumes power, finer grouping is not automatically better.

5.1 Why the smallest group may not be the lowest-power choice

Per-bit gating minimizes the number of flip-flops clocked for a given set of changing bits, but it also requires the greatest number of independent ICGs. Each additional ICG adds clock-pin capacitance, internal switching, leakage, enable circuitry, and clock routing. For a small register, the overhead can offset or exceed the clock power saved. Grouped clock gating therefore seeks a balance between avoiding redundant clock pulses and keeping the gating overhead manageable [1], [2]. The best choice should be determined with measured total power under representative input activity, rather than assumed from group size alone.

6. Extension to 16-bit and 32-bit Registers

The same grouping principle can be applied to wider register words. These tables provide structural estimates of the number of groups and two-input OR-gate equivalents. They are not transistor-level area estimates and do not predict absolute power in milliwatts. The estimates assume one XOR change detector per bit, one ICG per group, and binary OR reductions; a physical implementation may use different cells or optimize the logic.

For an N-bit register split into groups of k bits, where k divides N:

$$\text{Number of groups / ICGs} = N / k$$

$$\text{XOR change detectors} = N$$

$$\text{Two-input OR equivalents} = N - (N / k)$$

The OR-equivalent count follows because a k-input OR reduction can be built from k - 1 two-input OR gates. For k = 1, each bit directly controls its own ICG, so no inter-bit OR reduction is required.

6.1 Why absolute power is not extrapolated

It would be misleading to multiply the 8-bit readings by two or four and report the result as a 16-bit or 32-bit power estimate. Total power does not necessarily scale linearly with bit width because clock capacitance, data activity, grouping, routing, ICG overhead, supply conditions, and layout all affect the result. The wider-register tables therefore provide design-structure estimates only. Absolute power for 16-bit and 32-bit implementations must be measured in Microwind under stated, equivalent conditions.

7. Limitations and Experimental Requirements

The principal limitation is that only two power readings have been supplied. The conventional PIPO reference power is not yet available, so the study cannot report the DDCG power reduction relative to the conventional register. The exact power metric shown by Microwind, the process rule file, VDD, frequency, loading, input sequence, simulation window, and initial state must be documented to establish reproducibility.

The reported reduction between shared and grouped DDCG should be treated as a result for these two implementations and test conditions. Before asserting that grouping caused the difference, confirm that the conditions were identical and that the gated clocks are glitch-free. The probability model and wider-word tables are analytical aids, not measured outcomes. The paper does not claim measured area or timing performance.

Future Work and Recommendations:

Measure the conventional 8-bit PIPO power using the same exact input patterns and settings.

Verify and record the exact Microwind power metric and the .rul technology file.

Run low-group-only, high-group-only, both-group, and no-change test sequences.

Measure power for four 2-bit groups and per-bit gating if those circuits are implemented.

For 16-bit and 32-bit studies, measure each chosen group size instead of extrapolating milliwatts.

Store circuit screenshots, waveforms, and power result windows as evidence for the final report.

8. Conclusion

This paper focuses on low-power analysis of data-driven clock gating in an 8-bit PIPO register. The shared DDCG implementation has a supplied power reading of 188.336 mW, while the fixed 4-bit grouped implementation has a supplied reading of 175.422 mW. The grouped value is 12.914 mW lower, corresponding to a 6.8569% reduction relative to the shared reading. This comparison is specific to the two reported results and does not yet establish power improvement over a conventional PIPO register.

The analysis also explains the trade-off between one shared group, two 4-bit groups, four 2-bit groups, and individual per-bit gating. Smaller groups can reduce the number of flip-flops receiving clock pulses when data changes are sparse, but they require additional ICGs and control routing. The analytical extensions to 16-bit and 32-bit registers show how group counts and OR-reduction overhead change with width; they do not predict actual power in milliwatts. The next step is to complete controlled power measurements for the conventional baseline and other grouping choices. These measurements will allow a more reliable conclusion about which architecture offers the best low-power behavior for the tested circuit and input activity.

References

[1] S. Wimer and I. Koren, "Design Flow for Flip-Flop Grouping in Data-Driven Clock Gating," IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 22, no. 4, pp. 771-778, Apr. 2014, doi: 10.1109/TVLSI.2013.2253338. Available: https://doi.org/10.1109/TVLSI.2013.2253338

[2] S. Wimer, "On Optimal Flip-Flop Grouping for VLSI Power Minimization," Operations Research Letters, vol. 41, no. 5, pp. 486-489, Sep. 2013, doi: 10.1016/j.orl.2013.06.002. Available: https://doi.org/10.1016/j.orl.2013.06.002

[3] A. Ranganayakulu and K. Satyaprasad, "Subword Partition Based Data Driven Clock Gating Scheme for Low Power VLSI Design," International Journal of Computer Applications, vol. 108, no. 14, pp. 29-34, Dec. 2014, doi: 10.5120/18982-0419. Available: https://www.ijcaonline.org/archives/volume108/number14/18982-0419/

[4] Intel, "Recommended Clock-Gating Methods," Quartus Prime Pro Edition User Guide, Section 2.2.3.5.1, Apr. 14, 2025. Available: https://www.intel.com/content/www/us/en/docs/programmable/683082/25-1/recommended-clock-gating-methods.html

[5] J. M. Rabaey, A. Chandrakasan, and B. Nikolic, Digital Integrated Circuits: A Design Perspective, 2nd ed.